Assignment 2: Language Models and Author Identification

LING 5832 · Yiyuan Jia · September 22, 2026

Where everything is

All code and the full report are in a public GitHub repository:

https://github.com/yiyuan1840/ling-5832/tree/main/assignments/A2

deliverable	location
Report	assignments/A2/assignment2.pdf
Tokenizer	src/tokenizer.py
N gram models, add k smoothing, perplexity	src/ngram.py
Author identification	src/authorid.py
Corpus loading and splitting	src/corpus.py
Everything run end to end	main.py
The experiments behind every table in the report	src/experiments.py
Author ID labels for the test set	to follow once the test set is released

Running it

The project uses uv. From `assignments/A2`:

```
uv run main.py the whole pipeline uv run src/experiments.py every number in the report
```

Summary of results

A byte pair encoding tokenizer built on the Hugging Face `tokenizers` library, bigram and trigram models with add k smoothing, perplexity, and an author identification classifier trained on *The Hobbit* and *The Lost World*.

On held out 50 word passages the classifier is correct on 255 of 255. It still reaches 97.6% after removing the two easiest cues, the differing quote characters between the two editions and the character names, which is the more meaningful figure.

Three findings are discussed in the report: add 1 smoothing is nearly three times worse than k = 0.01; the trigram model loses to the bigram because it meets a context it has never seen 29.7% of the time against 0.2%; and per token perplexity is not comparable across vocabulary sizes, which makes the apparently best vocabulary the wrong choice.

Section 8 of the report describes which code I wrote and which was scaffolded by Claude (Anthropic), as the deliverable requires.